Koneru Lakshmaiah Education Foundation
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MACHINE LEARNING (25SC2107E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-3	Team No-15	Credit Card Fraud Detection

1. Abstract

Credit card fraud has become one of the fastest-growing financial crimes due to the rapid growth of online payments and digital banking. Detecting fraudulent transactions is challenging because fraud cases make up only a very small portion of all transactions, creating a highly imbalanced data set. Traditional rule-based systems often fail to identify new fraud patterns and may generate a high number of false alarms.

This project develops a Machine Learning-based Credit Card Fraud Detection System that detects suspicious transactions using both supervised and anomaly detection approaches. The system compares a supervised model trained with SMOTE (Synthetic Minority Oversampling Technique) against an unsupervised Isolation Forest model to evaluate their effectiveness in identifying fraudulent activities. The project focuses on improving fraud detection accuracy while minimizing false positives, making it suitable for real-world financial security applications.

2. Problem Statement

Financial institutions process millions of credit card transactions every day, making manual fraud detection impossible. Since fraudulent transactions represent only a tiny fraction of the total transactions, machine learning models often become biased toward predicting transactions as legitimate.

The objective of this project is to build an intelligent fraud detection system capable of accurately identifying fraudulent transactions in a highly imbalanced data set by comparing supervised and anomaly detection techniques while reducing false positives and improving detection performance.

3. Tools & Technologies

Tool	Purpose
Python	Programming language used for developing the fraud detection model.
Pandas	Data loading, cleaning, and reprocessing.
NumPy	Numerical computations and array operations.
Matplotlib	Creating graphs and visualizations.

Seaborn	Statistical data visualization and plotting.
Jupyter Notebook	Developing, testing, and documenting the project.
Git	Version control for tracking project changes.
GitHub	Hosting and sharing the project repository.

4. Team Details

Roll Number(s)	Name of the Student	Signature
2520030575	M. Likhitha	Likhitha
2520030198	K. Vaishnavi	Vaishnavi

Guide Name and Signature (remarks if any) _____